Signed number representation:

MSB → sign (0 → +ve, 1 → -ve)

non uniqueness of 0

1's complement:

+ve: binary

-ve: flip bits

non uniqueness of 0

2's complement:

+ve: binary

-ve: flip bits add 1

$-5+3 \Rightarrow 1011 + 0011 = 1110 \Rightarrow -2$

$5-(-2) \Rightarrow 0010 + 0101 = 0111 \Rightarrow 7$

Decimal to binary: divide by 2, take reminder until quotient is 0

for factions: $0.6875_{10} = 0.1011_2$

$0.6875 \times 2 = 1 + 0.3750$ $a_{-1} = 1$

$0.3750 \times 2 = 0 + 0.7500$ $a_{-2} = 0$

$0.7500 \times 2 = 1 + 0.5000$ $a_{-3} = 1$

$0.500 \times 2 = 1 + 0.0000$ $a_{-4} = 1$

Binary to hexadecimal: groups of 4

Binary to octal: groups of 3

Base n to decimal: $ABCD_n = D n^0 + C \cdot n^1 + B \cdot n^2 + A \cdot n^3$ to

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Axioms of boolean algebra:

$x+0=x$ $x \cdot 1 = x$ $x+y=y+x$ $x \cdot y = y \cdot x$

$x + x' = 1$ $x \cdot x' = 0$

Theorms: $x+x=x$; $x \cdot x = x$; $x+1=1$; $x \cdot 0 = 0$;

$(x')' = x$; $x+xy=x$; $(x+y)' = x'y'$; $(xy)' = x'+y'$

Duality: simply interchange and & or ; replace 1's by 0 & 0's by 1

Canonical form: Boolean functions expressed in terms of POS or SOP

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implicant: F of n-variables ; Product term P is implicant of F iff for every combination of values of variables for which $P=1$; F is also equal to 1

prime implicant: If any variable is deleted from an implicant , it'll no longe

be an implicant

essential prime implicants: a minterm covered by only one prime implicant

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universal gates: NAND, NOR

NAND for NOR:



NAND only network: AND output invert, OR input invert

NOR only network: AND input invert OR output invert

XOR gate: $a \oplus b = a\bar{b} + \bar{a}b$ odd no of 1's

XNOR gate: $a \odot b = ab + \bar{a}\bar{b}$ even no of 1's
